Hands-On-Activity No. 4.1

Hands-on Activity 4.1: Switch Case

Course Code: CPE 007	**Program: Computer Engineering**
Course Title: Programming logic and Design	**Date Performed: August 11, 2026**
Section: CPE11S2	**Date Submitted: August 11, 2026**
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1. Objective(s)

This activity aims to develop programming skills using switch case

2. Intended Learning Outcomes

2.1 Understand switch case control structure

2.2 Apply switch case structure in C++ programming

3. Materials and Equipment

Internet connection
Personal Computer

4. Output

1. Using switch statement Write a C++ program to input marks of five subjects Physics, Chemistry, Biology, Mathematics and Computer. Calculate percentage and grade according to following:
Percentage >= 90% : Grade A
Percentage >= 80% : Grade B
Percentage >= 70% : Grade C
Percentage >= 60% : Grade D
Percentage >= 40% : Grade E
Percentage < 40% : Grade F

_____

The output should be similar to the one below:
Physics grade: 80
Biology : 80
Math : 80
Computer : 80
Chemistry : 75

_____
Average is : 79.00
Grade Level: C

Answer:

```
#include <iostream>
#include <string>
using namespace std;

int main() {
   int Physics, Biology, Math, Computer, Chemistry;
   double g1, g2, g3, g4, g5;
   double sumGr;
      cout << "Enter Subject Grades:" << endl;
      cout << "**********************" << endl;
      cout << "Physics Grade: " << endl;
      cin >> g1;
```

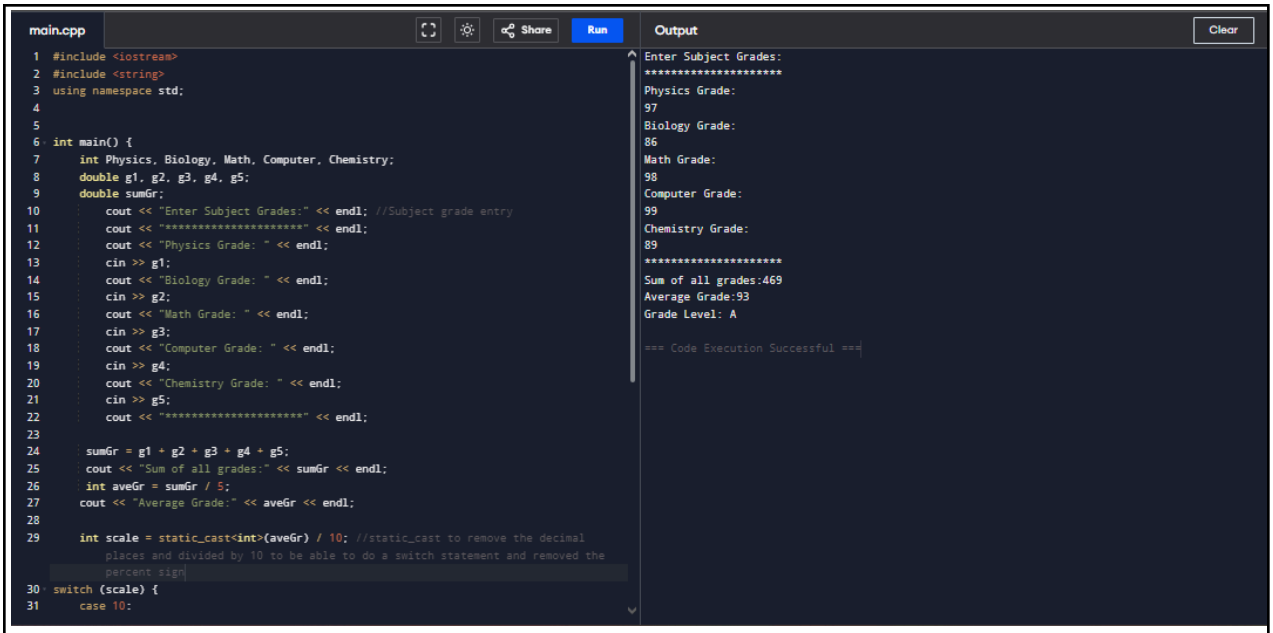
```
                cout << "Biology Grade: " << endl;
                cin >> g2;
                cout << "Math Grade: " << endl;
                cin >> g3;
                cout << "Computer Grade: " << endl;
                cin >> g4;
                cout << "Chemistry Grade: " << endl;
                cin >> g5;
                cout << "**********************" << endl;

             sumGr = g1 + g2 + g3 + g4 + g5;
             cout << "Sum of all grades:" << sumGr << endl;
             int aveGr = sumGr / 5;
            cout << "Average Grade:" << aveGr << endl;

            int scale = aveGr / 10;
            char FinalGrade;

          switch (scale) {
            case 10:
            case 9:
            cout << "Grade Level: A";
            break;
            case 8:
            cout << "Grade Level: B";
            break;
            case 7:
            cout << "Grade Level: B";
            break;
            case 6:
            cout << "Grade Level: B";
            break;
            case 5:
            case 4:
            cout << "Grade Level: B";
            break;
            default:
            cout << "Grade Level: F";
            break;
            }
            return 0;
          }
```

ScreenShot:



2.***Convert the above given problem into a solution using a Nested If structure in C++.***

Answer:

```
#include <iostream>
#include <string>
using namespace std;

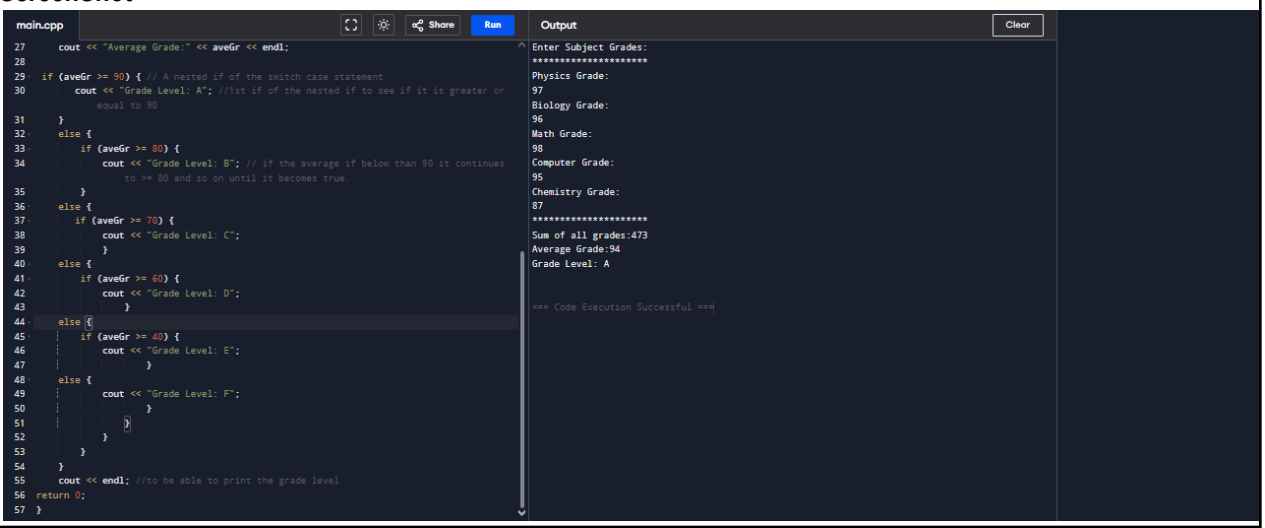

int main() {
   int Physics, Biology, Math, Computer, Chemistry;
   double g1, g2, g3, g4, g5;
   double sumGr;
      cout << "Enter Subject Grades:" << endl; //Subject grade entry
      cout << "**********" << endl;
      cout << "Physics Grade: " << endl;
      cin >> g1;
      cout << "Biology Grade: " << endl;
      cin >> g2;
      cout << "Math Grade: " << endl;
      cin >> g3;
      cout << "Computer Grade: " << endl;
      cin >> g4;
      cout << "Chemistry Grade: " << endl;
      cin >> g5;
      cout << "**********" << endl;

    sumGr = g1 + g2 + g3 + g4 + g5;
    cout << "Sum of all grades:" << sumGr << endl;
    int aveGr = sumGr / 5;
   cout << "Average Grade:" << aveGr << endl;

 if (aveGr >= 90) { // A nested if of the switch case statement
```

```
    cout << "Grade Level: A"; //1st if of the nested if to see if it is greater or equal to 90
  }
  else {
    if (aveGr >= 80) {
       cout << "Grade Level: B"; // if the average if below than 90 it continues to >= 80 and so on until it
becomes true.
    }
  else {
    if (aveGr >= 70) {
       cout << "Grade Level: C";
       }
  else {
    if (aveGr >= 60) {
       cout << "Grade Level: D";
          }
  else {
    if (aveGr >= 40) {
       cout << "Grade Level: E";
            }
  else {
       cout << "Grade Level: F";
            }
          }
       }
    }
  }
  cout << endl; //to be able to print the grade level
return 0;
}
```

ScreenShot



5. Conclusion

Learned about switch case programming.

6. Rubric